COMMUNITY-DETECTION PLATFORM

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ABSTRACT :

The Community-Detection Platform is a web-based application designed to identify and connect people based on their common interests, activities, skills, and preferences. The platform allows users to create profiles, join communities, discover groups, share posts, and communicate with other members having similar interests. Unlike traditional social-media platforms, the main focus of this system is to help users find relevant communities and build meaningful connections. The system provides features such as user registration, login, profile management, community creation, community discovery, post sharing, commenting, liking, and member interaction. A community detection mechanism analyzes user interests and activities to suggest suitable communities. Administrators can manage users, communities, posts, and reported content through an admin dashboard.The platform can be developed using technologies such as HTML, CSS, JavaScript, React.js, Node.js, Express.js, and MongoDB. The proposed system provides a simple, scalable, and user-friendly environment for discovering and participating in online communities.

INTRODUCTION :

In today's digital world, people use online platforms to communicate, share information, and find others with similar interests. However, many social platforms contain a large amount of information, making it difficult for users to discover communities that are actually relevant to them.The Community-Detection Platform is developed to solve this problem by providing a centralized platform where users can discover and participate in communities according to their interests. Users can create an account, enter their interests, search for communities, join groups, and interact with other members.The system can analyze information such as user interests, community topics, posts, and interactions to identify suitable communities. For example, a user interested in photography can be recommended photography communities, while a user interested in programming can discover programming-related groups.The platform also provides an administrator interface for managing users, communities, posts, and inappropriate content. This makes the system useful for educational institutions, organizations, hobby groups, and general online communities.The primary goal of this project is to create a simple and effective platform that improves community discovery and encourages meaningful interaction between users.

ALGORITHMS USED

- **Louvain Algorithm –** Detects communities by analyzing connections between users.
- Girvan–Newman Algorithm – Finds communities by removing important connecting links between groups.
- **Label Propagation Algorithm –** Groups users based on the labels of their connected users.
- **K-Means Clustering –** Groups users based on similar interests and activities.
- **BFS (Breadth-First Search) –** Explores user connections level by level.
- **DFS (Depth-First Search) –** Traverses connected users deeply to identify groups.
- **Cosine Similarity –** Measures the similarity between users based on their interests or activities.
- **Graph-Based Analysis –** Represents users as nodes and their relationships as edges to detect communities.

SOFTWARE REQUIREMENTS :

OPERATING SYSTEM :

- WINDOWS 10/11
- LINUX
- MACOS

FRONT-END TECHNOLOGIES :

- HTML5
- CSS3
- JAVASCRIPT
- REACT.JS

BACK-END TECHNOLOGIES :

- NODE.JS
- EXPRESS.JS

DATABASE:

- MONGODB

DEVELOPMENT TOOLS :

- VISUAL STUDIO CODE
- GIT AND GITHUB
- WEB BROWSER SUCH AS GOOGLE CHROME OR MICROSOFT EDGE

HARDWARE REQUIREMENTS :

MINIMUM HARDWARE REQUIREMENTS :

Component	Requirement
Processor	Intel Core i3 or equivalent
RAM	4 GB
Storage	128 GB
Display	1366 × 768 resolution
Display	Wi-Fi or Ethernet
Display	Standard keyboard
Display	Standard mouse

RECOMMENDED HARDWARE REQUIREMENTS :

Component	Requirement
Processor	Intel Core i5 or equivalent
RAM	8 GB or higher
Storage	256 GB SSD or higher
Display	Full HD
Network	Stable broadband/Wi-Fi
GPU	Integrated graphics is sufficient

LITERATURE SURVEY :

Community detection is an important area of social-network analysis that focuses on identifying groups of users with similar interests or strong relationships. Several algorithms have been developed for detecting communities in networks, including Louvain, Label Propagation, Infomap, and Walktrap. The Louvain method is widely used because it can efficiently identify communities in large networks. Label Propagation identifies communities by analyzing the labels of connected users. Existing social-media platforms also use user interests and interactions to recommend groups and content. These existing techniques provide a foundation for developing an effective Community-Detection Platform.

METHODOLOGY :

The proposed system begins with user registration and profile creation, where users provide their basic information and interests. The system collects information such as user interests, joined communities, posts, likes, comments, and connections. This information is analyzed to identify similarities between users. A community-detection technique such as Louvain or Label Propagation can be used to identify groups of related users. Based on the detected communities and user interests, the system recommends suitable communities. Users can then join communities, interact with members, share content, and participate in discussions.

THANK YOU